Tasnim Jahan

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Website — GitHub — Google Scholar — LinkedIn

Research Interests

• Computer Vision • Representation Learning • Few-Shot Learning • Domain Adaptation • Image Segmentation • Medical Image Analysis

Education

M.Sc. in Computer Science and Engineering (Data Science), CGPA: 3.92 / 4.00 Oct 2021 – Jun 2024
United International University *Dhaka, Bangladesh*
Summa Cum Laude (Highest Distinction), CGPA: 4.00 / 4.00 in last 5 semesters

M.Sc. in Mathematics, CGPA: 3.77 / 4.00 Jan 2016 – Apr 2017
Comilla University *Cumilla, Bangladesh*
Thesis: Bernoulli's Equation with the Variation of Velocity, Density, and Diameter of Pipe; Advisor: Zillur Rahman

B.Sc. in Mathematics, CGPA: 3.61 / 4.00 Mar 2010 – Jan 2016
Comilla University *Cumilla, Bangladesh*
Project: Numerical Study for Solving Initial Value Problems in Ordinary Differential Equations Using Single-Step and Multi-Step Methods; Advisor: Prof. Dr. Khalifa Mohammad Helal

Higher Secondary Certificate (HSC), Science, GPA: 5.00 / 5.00 Jul 2006 – Sep 2008
Mvi. Shamsul Karim College *Feni, Bangladesh*

Secondary School Certificate (SSC), Science, GPA: 5.00 / 5.00 Jan 2004 – Jun 2006
Chhagalnaiya Pilot Girls' High School *Feni, Bangladesh*

Publications

- **Tasnim Jahan**, Andrew P. King, Swakkhar Shatabda. HyProDINO: A Hybrid Prototype Guided DINO Framework for Cross-Domain Few-Shot Medical Image Segmentation. *17th International Workshop on Machine Learning in Medical Imaging (MLMI), MICCAI 2026.* **Selected for oral (18 of 66 accepted papers)** and poster presentation.
- **Tasnim Jahan**, Md Easin Arafat, Swakkhar Shatabda. ThyCLIPNet: A BiomedCLIP-Guided Lightweight Attention-Enhanced DeepLabV3+ Framework for Robust Thyroid Nodule Segmentation. *arXiv preprint arXiv:2610.04743*, 2026. Under revision at a Q1 journal.
- L. Al Amin, Md. Ismail Hossain, Thanh Thi Nguyen, **Tasnim Jahan**, Mahbubul Islam, Faisal Quader. Uncovering Critical Features for Deepfake Detection through the Lottery Ticket Hypothesis. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2025.

Research Experience

Independent Researcher (Computer Vision & Deep Learning) Jan 2024 – Present

Advisors: Prof. Andrew P. King (King's College London, UK); Prof. Swakkhar Shatabda (BRAC University, Bangladesh)

- **HyProDINO:** Developed a support-conditioned cross-domain few-shot medical image segmentation framework combining a trainable ResNet34–Transformer encoder, frozen DINOv2 semantic features, and prototype-guided support–query adaptation; evaluated on seven unseen target datasets, outperforming representative few-shot baselines in 5-shot evaluation. Ablation studies demonstrated absolute Dice improvements of 15.04% and 12.87% from DINOv2 integration on visually distant Chest X-ray and ISIC2018 domains, respectively. **(MLMI, MICCAI 2026)**.
- **Ongoing Research:** Extending HyProDINO through methodological development and cross-domain evaluation, focusing on improved representation learning and adaptation strategies. **(Journal manuscript in preparation)**.
- **ThyCLIPNet:** Developed lightweight encoder–decoder framework integrating BiomedCLIP-derived visual semantic guidance with attention-enhanced multi-scale CNN features for thyroid ultrasound segmentation; achieved Dice scores of 96.22%, 87.58%, 84.73%, and 80.70% on TG3K, TN3K, DDTI, and PKTN, respectively, with 8.55M parameters and 22.99 GFLOPs. **(Journal manuscript under revision)**.
- **Research Mentorship:** Selected as a mentee for the **RISE-MICCAI Mentorship Program (Oct 2025–Oct 2026)** under Prof. Andrew P. King, supporting an ongoing research collaboration.
- **Deepfake Detection:** Investigated Lottery Ticket Hypothesis-based iterative pruning across MesoNet, CNN-5, and ResNet-18 on OpenForensics and FaceForensics++; identified sparse winning-ticket subnetworks that preserved substantial detection performance at high sparsity and demonstrated cross-dataset transferability.**(IEEE SMC 2025)**.

Honors & Awards

MICCAI Society RISE Travel Grant — Selected recipient; up to $2,500 in travel support to attend MICCAI Aug 2026

Ranked 5th in LISA 2025 Challenge among international participants, RISE-MICCAI Summer School May 2025 – Oct 2025

Merit-based Scholarship (5 times) at United International University for academic performance Jan 2023 – Aug 2024

Projects

3D Hippocampus Segmentation (Ultra-low-field MRI) | *Python, MONAI, PyTorch* | Code Jul 2025 – Sep 2025

- Developed a MONAI-based 3D deep learning pipeline for bilateral hippocampus segmentation from ultra-low-field MRI.
- Implemented and compared Attention U-Net, DynUNet, SegResNet, and UNETR using class-balanced cropping and sliding-window inference.
- Among the tested models, Attention U-Net achieved the highest validation Dice (≈ 0.60) with post-processing.

Thyroid Nodule Segmentation | *Python, TensorFlow/Keras* | Code — Report Nov 2023 – Feb 2024

- Developed an end-to-end deep learning pipeline for thyroid nodule segmentation on the TN3K ultrasound dataset.
- Implemented and compared U-Net variants with pretrained encoders (VGG16, VGG19, ResNet50, MobileNetV2).
- Best among the tested encoders: MobileNetV2-based U-Net (Dice: 84.70%, IoU: 73.80%), achieving strong accuracy with a lightweight backbone.

Spatio-Temporal Data Analysis Using Deep Learning (Survey Paper) | *Literature Review* | ReportJul 2023 – Nov 2023

- Conducted a structured survey of deep learning methods for spatio-temporal data: data types, pipelines, and models.
- Covered CNNs, RNNs, GNNs, and Transformers for prediction, classification, clustering, and anomaly detection.
- Summarized real-world applications, open challenges, and research directions.

Hypo-thyroid Disease Prediction | *Python, Scikit-learn* | Code — Report Feb 2022 – May 2022

- Developed a machine learning pipeline for early detection of hypothyroid disorder using clinical and demographic data.
- Applied preprocessing techniques including imputation, feature encoding, scaling, and SMOTE for class imbalance.
- Trained and compared Decision Tree, Logistic Regression, SVM, and Random Forest classifiers.
- Achieved the best performance with Random Forest (Accuracy: 98.70%), outperforming other models.

Face Mask Detection | *Python, TensorFlow, OpenCV* | Code — Report Mar 2023 – Jul 2023

- Implemented a full pipeline for mask vs. no-mask detection using a MobileNetV2-based CNN.
- Included training, evaluation, and real-time inference with OpenCV.

Predict Customer's Future Purchases | *Python, Machine Learning* | Code — Report Jun 2022 – Sep 2022

- Developed a machine learning pipeline to predict customer purchases within 90 days using transactional retail data.
- Trained and evaluated Logistic Regression, Decision Tree, Random Forest, and Gaussian Naïve Bayes.
- Achieved the best performance with Logistic Regression (Accuracy: 99.10%, F1-score: 98.50%), while GaussianNB provided the fastest inference.

Bangla Financial Sentiment Classification | *Python, ML* | Code — Report Jan 2023 – Aug 2023

- Collected and labelled 1.8K+ Bangla financial headlines from the Prothom Alo newspaper for sentiment classification.
- Achieved the best performance with SVM (F1: 78%) and Logistic Regression (Accuracy: 76.60%).

LLM-Enhanced Stock Movement Prediction | *ML* | Report Aug 2023 – Sep 2023

- Integrated financial time-series data with information extracted from news using LLMs for weekly stock prediction.
- Improved Logistic Regression (Accuracy: 54.80% $\rightarrow$ 63%, F1: 42.10% $\rightarrow$ 60.90%).

IoT-Based Watering Automation and Security System for Plants & Pets | *Sensors* | Report Jul 2022 – Nov 2022

- Designed an IoT concept for automated watering with basic security monitoring for plants and pets.
- Focused on automation logic, sensor-driven triggers, and system-level design.

Skills

Programming: Python, SQL, Java, C, C++
Deep Learning: PyTorch, TensorFlow, Keras, MONAI
Computer Vision / ML: OpenCV, scikit-learn, NumPy, Pandas
Research Tools: LaTeX, Kaggle, Google Colab, Jupyter Notebook
Visualization: Matplotlib, Power BI
Soft Skills: Complex problem solving, rapid skill acquisition, effective communication

Relevant Coursework

- Advanced Artificial Intelligence
- Deep Learning
- Machine Learning
- Data Mining
- Natural Language Processing
- Research Methodology
- Advanced Database Systems
- Differential & Integral Equations
- Discrete Mathematics
- Linear Algebra
- Numerical Analysis
- Ordinary & Partial Differential Equations
- Methods of Statistics
- Vector Calculus & Tensor Analysis

Academic Service

- **Journal Reviewer**, Biomedical Signal Processing and Control (Q1) — reviewed 6 manuscripts.
- **Workshop Reviewer**, MICCAI 2026 MSB EMERGE and FAIMI–BRIDGE–EPIMI — reviewed 3 manuscripts.

Teaching Experience

Section Leader (Volunteer) Apr 2026 – May 2026
Stanford University – Code in Place *Remote*
- Selected through a competitive process from 2,000 applicants to join a global teaching team.
- Taught Python programming to a diverse cohort in a large-scale online course.
- Led problem-solving sessions and mentored students to improve programming and analytical skills.

Instructor – Mathematics Jan 2017 – Mar 2018
Institute of Computer Science and Technology (ICST) *Feni, Bangladesh*
- Taught mathematics to 200+ diploma-level students across engineering disciplines.
- Designed lectures, assignments, and assessments tailored to diverse learning needs.

Junior Teacher – Mathematics Mar 2018 – Dec 2018
Mainamati English School and College *Cumilla, Bangladesh*
- Delivered mathematics instruction in English medium and monitored student performance.
- Founded and led a math club to improve student engagement.

Professional Experience

Freelance Data Analyst Jan 2023 – Feb 2024
Fiverr *Remote*
- Developed interactive dashboards and data-driven reports using Microsoft Power BI.
- Performed data cleaning, statistical analysis, and feature engineering on real-world datasets.

Field Data Collector Nov 2021 – Dec 2021
BRAC James P Grant School of Public Health *Dhaka, Bangladesh*
- Collected qualitative and quantitative data from ∼100 participants using Kobo Toolbox.
- Ensured data quality, consistency, and proper documentation for analysis.

Programme Organizer – Education Jan 2022 – Apr 2022
Jagorani Chakra Foundation *Cox's Bazar, Bangladesh*
- Supported the implementation of a UNICEF-funded education program by training teachers in effective teaching methodologies and classroom practices.
- Collaborated with stakeholders, monitored field activities, and contributed to reporting and documentation.

Education Facilitator – Education Jan 2019 – Apr 2020
DanChurch Aid *Cox's Bazar, Bangladesh*
- Delivered literacy and numeracy sessions to ∼300 learners in a UNICEF-funded program.
- Designed course materials and conducted baseline and endline data collection.

Online Courses & Certifications

WorldQuant University: Computer Vision Lab — Dec 2025 – Present

RISE-MICCAI Summer School (MICCAI Society) — Jul 2025

Google: Data Analytics Specialization — Apr 2021 – Jun 2021

University of Amsterdam: Basic Statistics — Jun 2021 – Jul 2021

Extracurricular Activities

English & Bangla News Presenter at Bangladesh Betar, Cox's Bazar, Bangladesh — Apr 2025 – Present

Member at Partibartan, Cultural Organization, Comilla University, Cumilla, Bangladesh — Jan 2014 – Jan 2017